6COSC022W Coursework 1 (2025/26)	
Module leader	Indrajith Ekanayake
Unit	Coursework 1
Weighting:	40%
Qualifying mark	30%
Description	Design of web application and essay
Learning Outcomes Covered in this Assignment:	L02. LO3, LO4
Handed Out:	
Due Date	**2nd April 2026, 11.59pm IST**
Expected deliverables	Electronic document and web application deployed locally
Method of Submission:	online via Blackboard
Type of Feedback and Due Date:	Feedback will be given during viva following deadline and via Blackboard rubric. **All marks will remain provisional until formally agreed by an Assessment Board.**

Assessment regulations

Refer to section 4 of the “How you study” guide for undergraduate students for a clarification of how you are assessed, penalties and late submissions, what constitutes plagiarism etc.

Penalty for Late Submission

If you submit your coursework late but within 24 hours or one working day of the specified deadline, 10 marks will be deducted from the final mark, as a penalty for late submission, except for work which obtains a mark in the range 40 – 49%, in which case the mark will be capped at the pass mark (40%). If you submit your coursework more than 24 hours or more than one working day after the specified deadline you will be given a mark of zero for the work in question unless a claim of Mitigating Circumstances has been submitted and accepted as valid.

It is recognized that on occasion, illness or a personal crisis can mean that you fail to submit a piece of work on time. In such cases you must inform the Campus Office in writing on a mitigating circumstances form, giving the reason for your late or non-submission. You must provide relevant documentary evidence with the form. This information will be reported to the relevant Assessment Board that will decide whether the mark of zero shall stand. For more detailed information regarding University Assessment Regulations, please refer to the following website:http://www.westminster.ac.uk/study/current-students/resources/academic-regulations

Coursework Description

Alumni Influencers - Web API Development Assignment

Background Scenario

Phantasmagoria Ltd, a London based educational technology company, has been commissioned by the University of Eastminster to revolutionise alumni engagement through augmented reality and a new influencer platform.

The Problem

The current system is a static webpage listing alumni. This approach has several disadvantages: Low student engagement, Outdated information, Limited alumni details (current roles unknown), Minimal interaction between students and graduates

The Solution

Phantasmagoria's platform transforms alumni engagement into a self sustaining marketplace that works as follows:

How It Works

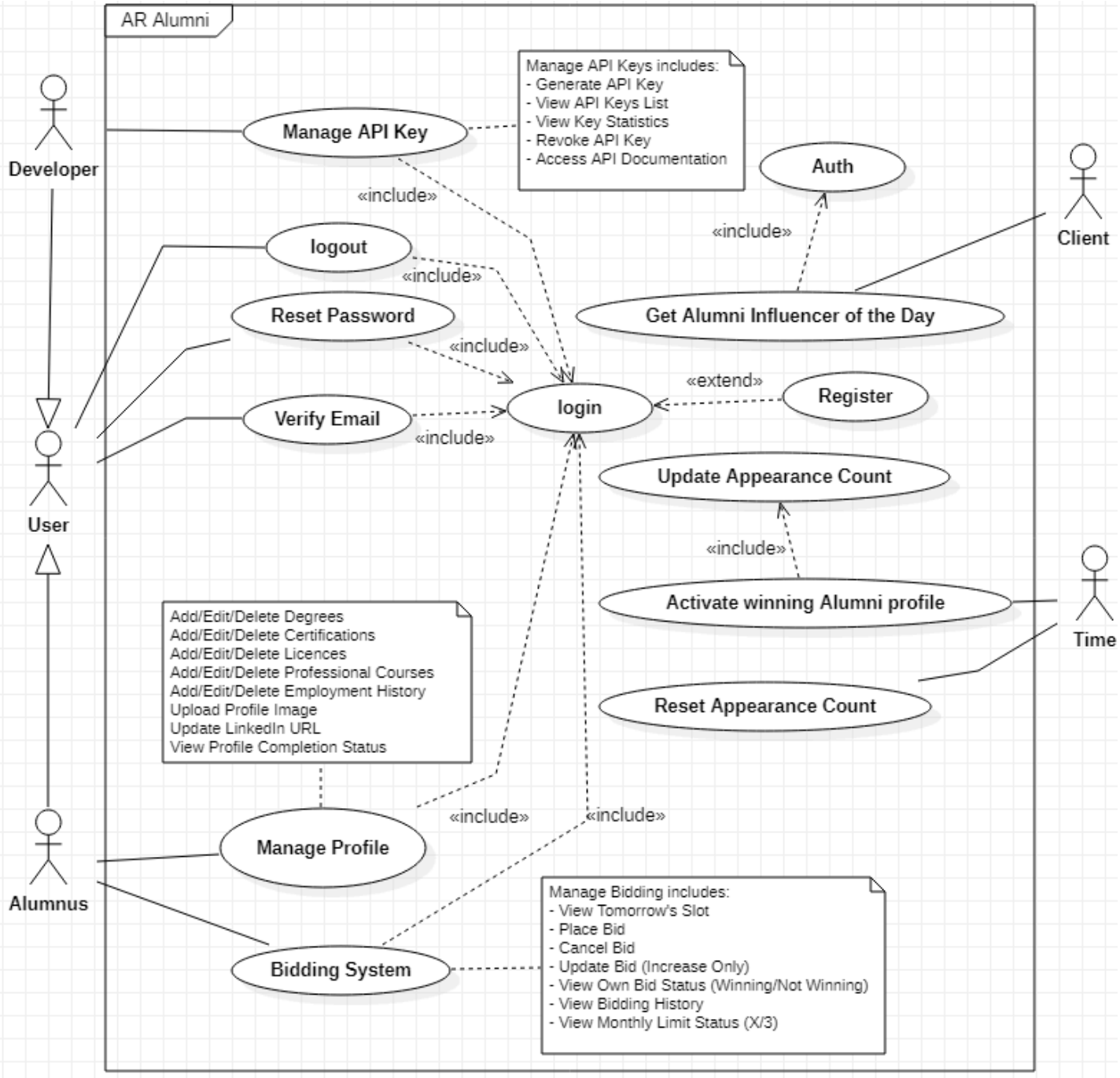
1. Alumni as Influencers: Successful alumni with strong LinkedIn profiles, professional courses, licences, and certifications leverage their credibility on the platform.
2. Sponsorship Model: Professional course providers, licensing bodies, and certification organisations sponsor alumni to promote their offerings similar to how brands sponsor social media influencers. Sponsors only pay the influencer if the influencer wins and their profile is displayed. If user A has 4 certifications. 2 of them want to be endorsed and so offer user A, £200 and £300 respectively. User A may then bid £250 and if they win, they get to pocket £250.
3. Daily Bidding System: Alumni use sponsorship money to bid for exclusive 24 hour featured slots on the platform. Each day at 6 PM, the highest bidder wins and their profile becomes the exclusive Alumni of the Day for the entire following day, visible to all students.
4. Benefits for Students: Students see relatable success stories from people who walked the same halls, learn about career applications of their degree, connect via LinkedIn, and are inspired to build their own professional profiles for future participation.

Bidding Rules

- Blind Bidding: Alumni cannot see the current highest bid but receive feedback on whether they're winning or losing. This ensures fair competition while maintaining strategic privacy.
- Diversity Limits: Each alumnus can be Alumni of the Day a maximum of 3 times per calendar month. However, participating in one university alumni event grants them the opportunity to bid for a 4th time that month.

Your Task is to build a web based API to serve up the data to the AR client. You need not concern yourselves with the functionality of the client, the API should be client agnostic.

Client Sample here


AR Alumni
Developer
User
Alumnus
Client
Time
Manage API Key
Manage API Keys includes:
- Generate API Key
- View API Keys List
- View Key Statistics
- Revoke API Key
- Access API Documentation
Auth
«include»
logout
Reset Password
Verify Email
login
Get Alumni Influencer of the Day
«extend»
Register
Update Appearance Count
Activate winning Alumni profile
Reset Appearance Count
Add/Edit/Delete Degrees
Add/Edit/Delete Certifications
Add/Edit/Delete Licences
Add/Edit/Delete Professional Courses
Add/Edit/Delete Employment History
Upload Profile Image
Update LinkedIn URL
View Profile Completion Status
Manage Profile
Bidding System
Manage Bidding includes:
- View Tomorrow's Slot
- Place Bid
- Cancel Bid
- Update Bid (Increase Only)
- View Own Bid Status (Winning/Not Winning)
- View Bidding History
- View Monthly Limit Status (X/3)

Part 1 Features Overview

Core Features You Will Build

1. **Alumni Registration & Authentication**
    - Email-based registration (university domain required)
    - Email verification system
    - Secure login/logout with sessions
    - Password reset functionality
2. **Complete Alumni Profile Creation with web pages**
    - Personal information and biography
    - LinkedIn profile URL
    - Degrees (with URLs to official university degree pages) & completion date
    - Professional certifications (with URLs course pages) & completion date
    - Professional licences (with URLs to licence awarding body) & completion date
    - Short professional courses (with URLs to course pages) & completion date
    - Employment history start and end dates
    - Profile image upload
3. **Blind Bidding System**
    - Place bids without seeing highest bid amount
    - Receive win/lose status feedback
    - Update bids (increase only)
    - Monthly limit enforcement (3 features per month)
    - Automated midnight winner selection (Mark the profile that will be displayed)
4. **Security to restrict client Access through bearer tokens**
    - Decide how different clients will access the data
    - View usage statistics i.e. number of times and time stamps of clients logging in and usage of keys / tokens and endpoints accessed
    - Ability to revoke security tokens
5. **Public Developer API**
    - Get today's featured alumnus
    - Comprehensive Swagger documentation

Alumni Registration & Authentication (15 marks)

Email Registration System (4 marks)

- Excellent (4): Complete registration with domain validation, strong password requirements, duplicate checking, proper error handling
- Good (2-3): Functional registration with minor validation issues
- Fair (1): Basic registration with significant validation gaps
- Poor (0): Non-functional or insecure registration

Email Verification System (4 marks)

- Excellent (4): Complete email verification with secure tokens, expiry handling, prevents unverified login
- Good (2-3): Working verification with minor security concerns
- Fair (1): Basic verification with security issues
- Poor (0): Missing or non-functional verification

Login/Logout & Session Management (4 marks)

- Excellent (4): Secure session management, proper authentication, timeout handling, secure logout
- Good (2-3): Working authentication with minor session issues
- Fair (1): Basic login with session security concerns
- Poor (0): Insecure or non-functional authentication

Password Reset Functionality (3 marks)

- Excellent (3): Complete password reset flow with secure tokens, expiry, email sending
- Good (2): Working reset with minor security gaps
- Fair (1): Basic reset with security concerns
- Poor (0): Missing or insecure password reset

Complete Alumni Profile Creation (5 marks)

Profile Management System (5 marks)

- Excellent (5): All required sections implemented with web forms, URL validation, multiple entry support, image upload, edit/delete functionality
- Good (3-4): Most profile sections working with minor issues
- Fair (1-2): Basic profile with missing sections or limited functionality
- Poor (0): Incomplete or non-functional profile system

Blind Bidding System (8 marks)

Bid Placement & Updates (2 marks)

- Excellent (2): Complete bid placement without revealing highest bid, increase-only updates, proper validation
- Good (1): Working bid system with minor blind bidding issues
- Poor (0): Reveals highest bid or non-functional bidding

Bid Status & Feedback (2 marks)

- Excellent (2): Complete bid tracking with status display and email notifications
- Good (1): Basic tracking with missing features
- Poor (0): Missing status tracking or notifications

Monthly Limit Enforcement (2 marks)

- Excellent (2): Accurate monthly tracking, 3-win limit enforced, displays remaining slots
- Good (1): Basic tracking with enforcement issues
- Poor (0): Inaccurate tracking or limit not enforced

Automated Winner Selection (2 marks)

- Excellent (2): Automated midnight selection, correct highest bid logic, profile marking, notifications
- Good (1): Working automation with minor issues
- Poor (0): Manual selection or incorrect automation

Security Implementation (20 marks)

Password Security (5 marks)

- Excellent (5): Bcrypt hashing with appropriate salt rounds, password strength validation, secure password storage.
- Good (3-4): Password hashing implemented with minor security gaps
- Fair (1-2): Basic password hashing with security concerns
- Poor (0): Passwords stored in plain text or weak hashing

Input Validation & Sanitization (5 marks)

- Excellent (5): Comprehensive input validation on all forms, SQL/NoSQL injection prevention, XSS protection, data sanitization
- Good (3-4): Good input validation with minor gaps
- Fair (1-2): Basic validation with security vulnerabilities
- Poor (0): No input validation or sanitization

Authentication Token Security (5 marks)

- Excellent (5): Secure token generation (verification, password reset), cryptographically random, proper expiry, single-use tokens, API key scoping.
- Good (3-4): Token security implemented with minor issues
- Fair (1-2): Basic token implementation with security gaps
- Poor (0): Insecure or predictable tokens

Security Headers & Protection (5 marks)

- Excellent (5): Helmet.js or equivalent implemented, CORS properly configured, CSRF protection, rate limiting on sensitive endpoints
- Good (3-4): Most security headers and protections in place
- Fair (1-2): Basic security measures with gaps
- Poor (0): No security headers or protections

Data Management (7 marks)

Excellent (6-7 marks)

- Database in 3NF with proper normalization
- Appropriate security fields (password hashing, tokens)
- Bid tracking and monthly limit enforcement
- Well-structured relationships between collections
- Efficient queries and appropriate indexing

Good (4-5 marks)

- Good database structure with minor normalization issues
- Most security fields implemented correctly
- Basic bid tracking present
- Adequate relationships between collections

Fair (2-3 marks)

- Basic database structure with normalization gaps
- Some security fields missing or poorly implemented
- Limited bid tracking functionality
- Weak relationships between collections

Poor (0-1 marks)

- Poor database design
- Missing security fields

- No bid tracking
- Minimum required collections only (Users, Profiles, Bids)

Architecture & Documentation (15 marks)

Excellent (12-15 marks)

- Deliberate architecture design with clear component separation
- Complete .env.example file with all required environment variables documented
- Detailed code comments explaining complex logic and algorithms
- Full Swagger/OpenAPI documentation with interactive UI at /api-docs
- Complete API endpoint documentation with request/response examples
- Database schema documentation with entity relationships

Good (8-11 marks)

- Some architecture documentation present
- Environment variables listed but lacking detailed explanations
- Code comments present for most complex sections
- Basic Swagger documentation with most endpoints covered
- Some gaps in documentation completeness

Fair (4-7 marks)

- Missing or unclear architecture documentation
- Environment variables not properly documented
- Sparse code comments
- Incomplete Swagger documentation or missing endpoints
- Limited API documentation

Poor (0-3 marks)

- No architecture documentation or diagrams
- Environment variables not documented
- Missing code comments
- No Swagger documentation or API documentation
- Unable to set up and run application from provided documentation

Viva Presentation (30 marks)

Application Demonstration (10 marks)

- Excellent (9-10): Clear demonstration of all features, smooth execution, well-organized presentation
- Good (7-8): Good demonstration of most features with minor issues
- Fair (5-6): Basic demonstration with some technical problems
- Poor (0-4): Incomplete or unclear demonstration

Code Explanation (10 marks)

- Excellent (9-10): Clear explanation of code structure, design patterns, and implementation details
- Good (7-8): Good explanation of main code components
- Fair (5-6): Basic code explanation with some gaps
- Poor (0-4): Poor or incomplete code explanation

Security & Design Justification (10 marks)

- Excellent (9-10): Comprehensive explanation of security implementation choices, architectural decisions, performance considerations, and testing approach
- Good (7-8): Good explanation of most major decisions
- Fair (5-6): Basic explanation of some decisions
- Poor (0-4): Poor or missing justification

Deliverables:

1. Submit a link to your Github Repo but it is your responsibility to ensure it is correct and accessible by your marker.

2. Viva (Failure to show up to defend your work will result in a grade of Zero.)

Use of Generative AI is forbidden in this coursework.

If you need any clarification about any of these issues, please contact me by email (at indrajith.e@iit.ac.lk) or in the tutorials or make an appointment to see me in my office.

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